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PREDICTIVE MODELING OF ARRHYTHMIC COMPLICATIONS POST-AMI
Using interpretable and non-interpretable machine learning models

Abstract

Objective: To develop and evaluate machine learning models for predicting arrhythmic complications in patients hospitalized for acute myocardial infarction, using routinely collected clinical and ECG data.
Methods: A real-world dataset of 1,700 AMI patients with 111 variables was preprocessed through variance/correlation filtering, Winsorization, and imputation. Following SMOTE oversampling and RobustScaler standardization, seven classifiers (Logistic Regression, Decision Tree, KNN, SVM, Random Forest, Gradient Boosting, and ANN) were trained and validated via stratified 10-fold cross-validation. A soft-voting ensemble model was also tested. Feature relevance was explored using SHAP, LIME, and permutation importance.
Results: All models yielded comparable performance (F1: 0.36–0.41; AUC: 0.65–0.69), suggesting performance limitations due to data constraints rather than algorithmic complexity. Most predictive features included rare ECG anomalies, age, lidocaine use, and markers of metabolic and inflammatory imbalance.
Conclusion: Models showed moderate performance, with limited sensitivity and low precision, likely due to data quality constraints. Still, the consistent emergence of clinically relevant predictors supports the plausibility of the approach. While not suitable for clinical decision-making at this stage, these models may assist early risk stratification if further refined and validated.

Clinical Background

Arrhythmic complications after AMI—such as ventricular tachycardia, fibrillation, atrial fibrillation, and AV blocks—are frequent and potentially life-threatening. Ventricular arrhythmias are a major cause of sudden cardiac death, while AF and AV blocks can lead to heart failure and hemodynamic instability. Up to 80% of AMI patients may develop arrhythmias during hospitalization. Early risk stratification is essential to guide monitoring and therapeutic decisions. However, prediction remains difficult due to the multifactorial and infrequent nature of these events. This project investigates whether machine learning models can predict arrhythmic complications using routine clinical data, comparing interpretable and non-interpretable approaches in terms of both performance and explainability.

Methods

This study analyzed data from 1,700 patients admitted for AMI at the Krasnoyarsk Interdistrict Clinical Hospital. The dataset included 111 features spanning demographics, laboratory results, therapies, ECG characteristics, and medical history. For this study, we considered as positive outcome the occurrence of any arrhythmic complication, resulting in a binary classification problem. The dataset is highly imbalanced, with approximately 18% of patients experiencing an arrhythmic complication (Table 1).

Complication	Prevalence (%)
Atrial fibrillation	10.0%
Supraventricular tachycardia	1.2%
Ventricular tachycardia	2.5%
Ventricular fibrillation	4.2%
Atrioventricular block	3.4%

Table 1. Prevalence of arrhythmic complications during hospitalization
This table details the specific cardiac arrhythmias that comprise the composite binary outcome used in this study. Each condition was coded as present or absent for each patient, and the outcome variable was defined as positive if at least one arrhythmic complication occurred. The overall prevalence of arrhythmic events was approximately 18%.

Features with similar clinical significance and low variance were aggregated to reduce dimensionality and enhance model robustness (Figure 1).

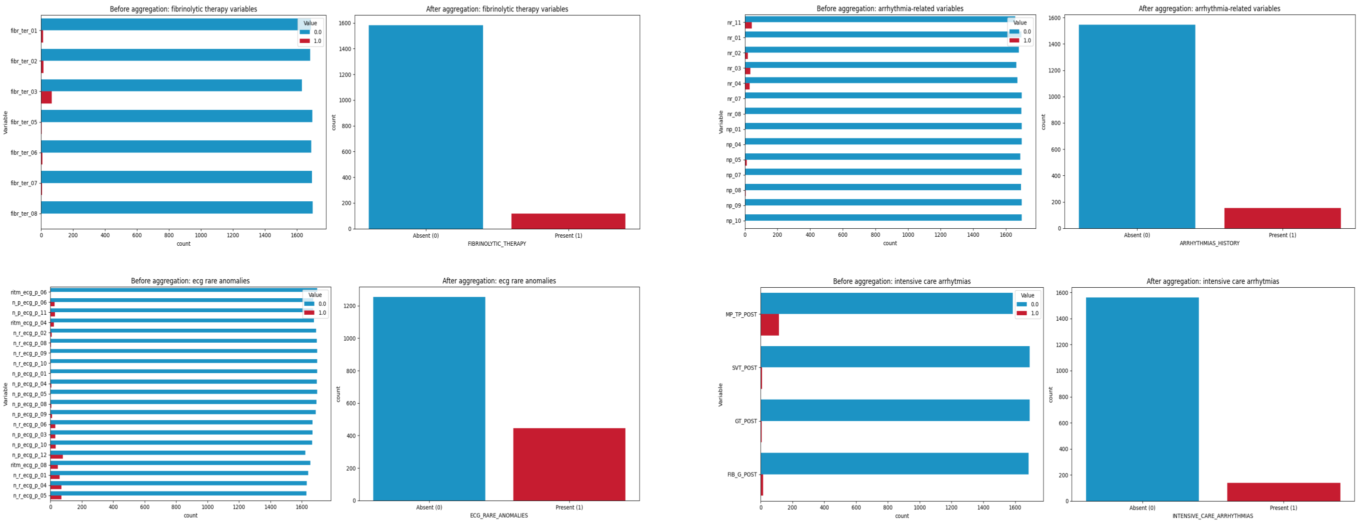


Figure 1. Aggregation of sparse and clinically related features
This figure shows how low-variance categorical features with overlapping clinical meaning were aggregated into composite variables to reduce dimensionality. Rare pre-existing arrhythmias were grouped into ARRHYTHMIAS_HISTORY, the use of fibrinolytic agents into FIBRINOLYTIC_THERAPY, uncommon ECG abnormalities at admission into ECG_RARE_ANOMALIES, and ICU arrhythmias into INTENSIVE_CARE_ARRHYTHMIAS.

In addition, several ordinal variables had high cardinality with sparse categories. To enhance model tractability, low-frequency or clinically redundant categories were merged based on medical reasoning rather than purely statistical criteria, preserving clinical relevance (e.g., Figure 2).

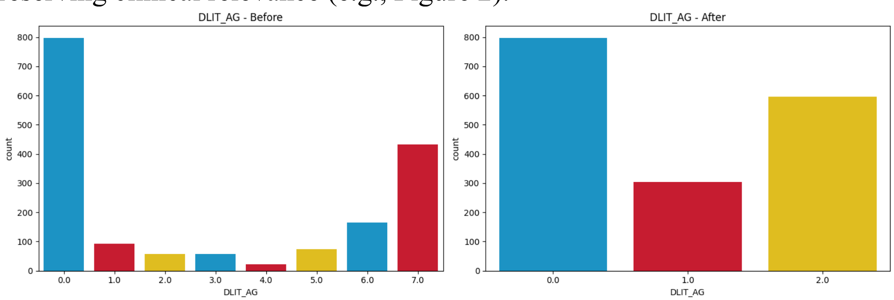


Figure 2. Ordinal feature aggregation
The variable DLIT_AG (duration of arterial hypertension) originally included eight categories, many with sparse representation. These were aggregated into three clinically meaningful levels: no hypertension (0), short duration (1–5 years), and prolonged duration (≥6 years). This aggregation reduced cardinality while preserving clinical relevance and improving model tractability.

In the exploratory phase, descriptive statistics were performed separately for numerical and categorical variables. Numerical variables were assessed for normality using the Shapiro-Wilk test. Distributions were summarized with median and interquartile range, and visualized through histograms, box plots, and pair plots. Categorical features were described using absolute and relative frequencies and visualized with bar plots. The dataset was also checked for duplicates, which were not found.

The data were split into training (80%) and testing (20%) sets using stratified sampling to preserve class distribution (Table 2).

Set	No Complication	Complication	Total
Train	1113	247	1360
Test	278	62	340
Overall	1391	309	1700

Table 2. Distribution of arrhythmic complications across training and test sets

All preprocessing steps—including imputation, scaling, feature selection, and SMOTE—were applied exclusively to the training set to avoid data leakage.

Missing values were handled by removing variables with more than 50% missingness and excluding patients with excessive missingness (>30 variables). Remaining missing values were imputed using the median for numerical variables and the mode for categorical ones (Figure 3).

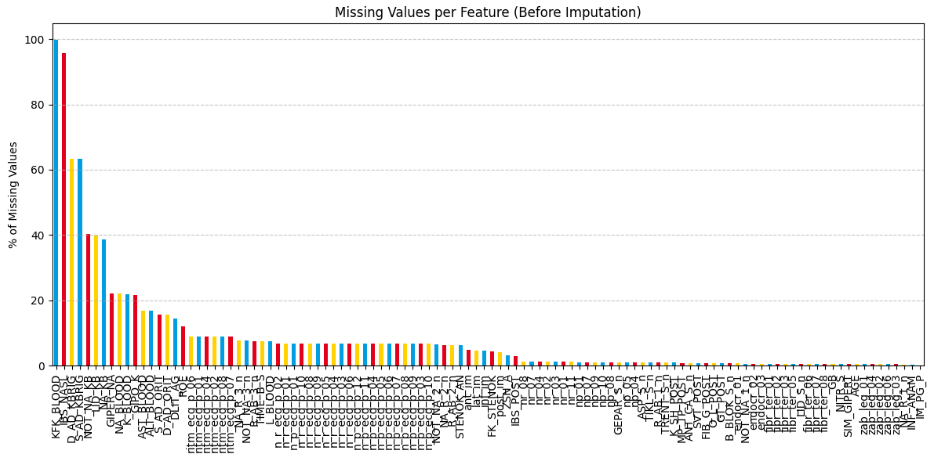


Figure 3. Percentage of missing values per feature before imputation

Outliers in numerical features were addressed using Winsorization based on interquartile range (IQR) with a 1.5-fold cap. Since all categorical variables were either binary or ordinal, no encoding was necessary. Numerical features were scaled using RobustScaler to reduce the impact of extreme values.

Feature selection was carried out in two steps:

1. Unsupervised filtering: Features with near-zero variance (threshold = 0.05) or with high pairwise correlation (Pearson $r > 0.7$) were removed.
2. Supervised selection: On the training set, we applied:
    - *Chi-squared test* for categorical features
    - *Linear Discriminant Analysis (LDA)* coefficients for numerical features

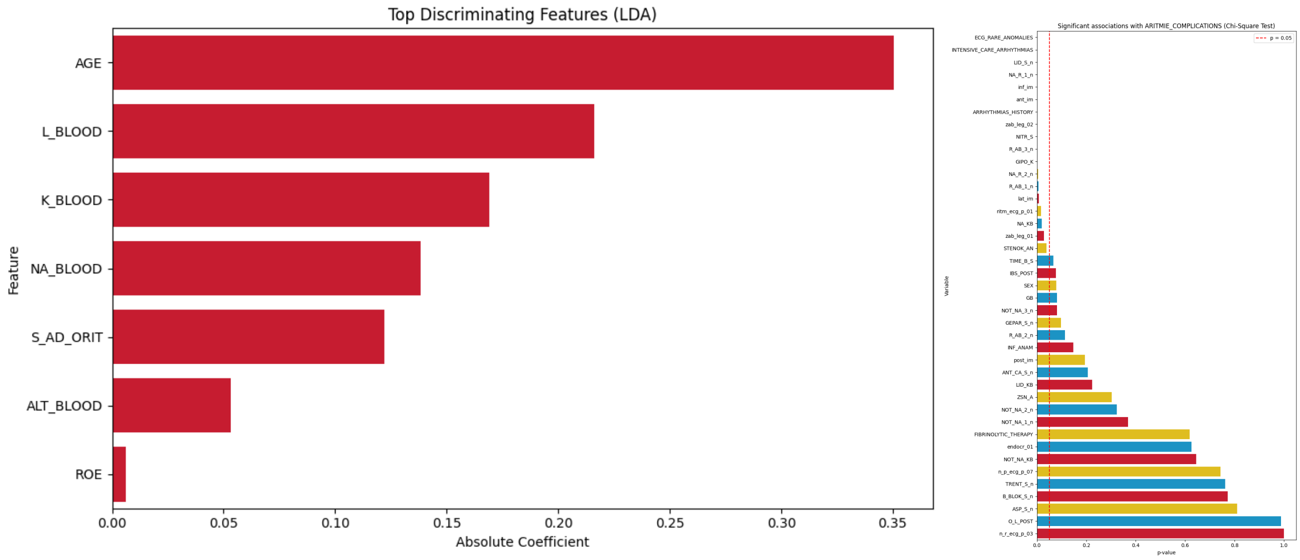


Figure 4. Supervised feature selection on training data

Left panel: The most discriminating numerical features identified via Linear Discriminant Analysis (LDA), ranked by the absolute value of their coefficients. ALT_BLOOD *and* ROE *were excluded due to low discriminative power.*

Right panel: Categorical features evaluated using the Chi-squared test. Only variables with $p < 0.05$ (to the left of the red line) were retained. Features with higher p-values were excluded from modeling to improve generalizability and reduce noise.

Seven machine learning models were trained: logistic regression, decision tree, K-nearest neighbors (KNN), support vector machine with linear kernel (SVM), random forest, gradient boosting (XGBoost), and a fully

connected artificial neural network (ANN). Hyperparameters were tuned using RandomizedSearchCV, and model performance was validated with 10-fold cross-validation, where SMOTE was applied only to the training folds to balance class representation (Table 3).
A soft voting ensemble combining all classifiers was also evaluated.

Model	Hyperparameters Explored	Hyperparameters Choosen
Logistic Regression	**C**: log-uniform [1e-5, 10]	**C**: 0.014077923139972392
Decision Tree	**max_depth**: [None, 3, 4, 5, 8, 10, 15, 20], **min_samples_split**: randint(2, 20), **min_samples_leaf**: randint(1, 30), **criterion**: [gini, entropy]	**max_depth**: 8 **min_samples_split**: 3 **min_samples_leaf**: 6 **criterion**: entropy
Random Forest	**n_estimators**: randint(100, 300), **max_depth**: [None, 4, 5, 8, 10, 20, 30], **min_samples_split**: randint(2, 10), **min_samples_leaf**: randint(1, 10), **max_features**: [sqrt, log2]	**n_estimators**: 161 **max_depth**: 4 **min_samples_split**: 8 **min_samples_leaf**: 7 **max_features**: sqrt
Gradient Boosting	**learning_rate**: uniform(0.01, 0.1), **n_estimators**: randint(100, 300), **Subsample**: uniform(0.5, 0.5), **max_depth**: randint(3, 6), **min_samples_split**: randint(4, 10), **min_samples_leaf**: randint(2, 10), **max_features**: [sqrt, log2]	**learning_rate**: 0.028182496720710062 **n_estimators**: 121 **Subsample**: 0.5035331526098588 **max_depth**: 3 **min_samples_split**: 5 **min_samples_leaf**: 5 **max_features**: sqrt
k-NN	**n_neighbors**: [3–50], **weights**: [uniform, distance], **p**: [1, 2]	**n_neighbors**: 31 **weights**: uniform **p**: 1
SVM (Linear)	**C**: log-uniform [1e-4, 10] **kernel**: linear **gamma**: [auto, scale]	**C**: 0.09846738873614563 **kernel**: linear **gamma**: scale
ANN (MLPClassifier)	**hidden_layer_sizes**: [(5,), (10,), (20,), (30,)] **activation**: [relu, tanh], **alpha**: uniform(0.0001, 0.01), **learning_rate_init**: uniform(0.0001, 0.001), **solver**: [adam], **early_stopping**: True	**hidden_layer_sizes**: (30,) **activation**: tanh **alpha**: 0.00024079822715084454 **learning_rate_init**: 0.0008068573438476172 **solver**: adam **early_stopping**: True

Table 3. Hyperparameter search space and selected values for each model
This table summarizes the hyperparameter tuning process performed via RandomizedSearchCV. For each model, the explored search space is reported along with the optimal configuration selected based on 10-fold cross-validation performance on the training set.

Model evaluation on the independent test set included accuracy, precision, recall, F1-score, AUC, ROC curves, and confusion matrices. Given the class imbalance, the F1 score was selected as the main evaluation metric to reflect the trade-off between precision and recall.

To improve model interpretability, we applied SHAP (SHapley Additive exPlanations) to examine both global and local feature importance, permutation feature importance to assess variable contribution, and LIME (Local Interpretable Model-Agnostic Explanations) to provide case-level explanations, supporting potential clinical application.

Results

All models were trained and evaluated on the task of predicting arrhythmic complications in AMI patients using a final set of 23 features. Performance was assessed on the test set (Table 4).

Explainability analyses were conducted using SHAP, permutation importance, and LIME (Figure 5-6). Across all models, a consistent pattern emerged: rare ECG anomalies, patient age, and lidocaine administration by the emergency team were the most influential predictors. These findings are clinically meaningful, as they reflect established risk factors for arrhythmias and early signs of electrical instability. SHAP provided global feature importance, while LIME visualized individualized predictions.

Model	Accuracy	AUC	Precision (1)	Recall (1)	F1-score (1)
Logistic Regression	0.64	0.68	0.27	0.58	**0.37**
Decision Tree	**0.76**	0.61	**0.33**	0.31	0.32
KNN	0.68	0.62	0.26	0.42	0.32
SVM (linear)	0.61	0.68	0.26	**0.60**	0.36
Random Forest	0.71	**0.70**	0.28	0.40	0.33
Gradient Boosting	0.75	0.68	0.32	0.32	0.32
ANN (MLP)	0.55	0.65	0.25	0.73	0.37

Table 4. Test set performance metrics for all models

This table reports the performance of all seven classifiers on the held-out test set. Despite differences in algorithmic complexity, all models achieved similar results, with F1-scores between 0.36 and 0.41 and AUC values ranging from 0.65 to 0.69. Metrics refer specifically to the positive class (patients with arrhythmic complications).

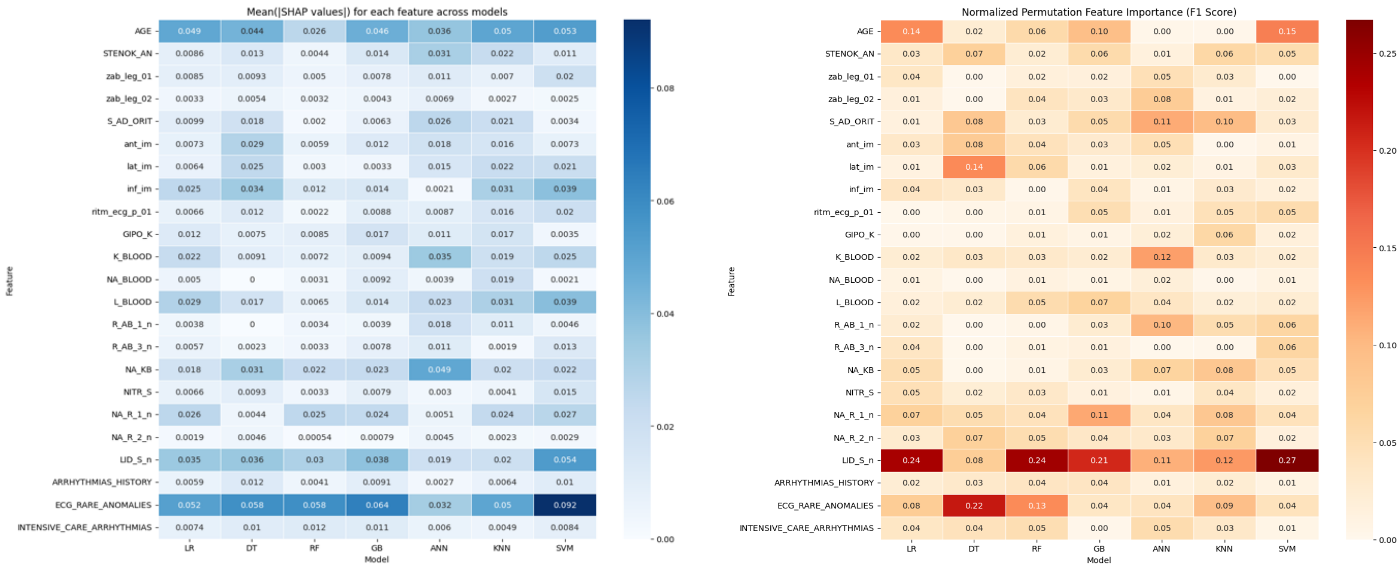


Figure 5. Global feature importance across models using SHAP and permutation analysis

Left: Mean SHAP values across models, showing the global impact of each feature on predictions.
Right: Permutation-based feature importance (normalized F1 impact). Rare ECG anomalies, age, and lidocaine use consistently ranked among the most predictive features.

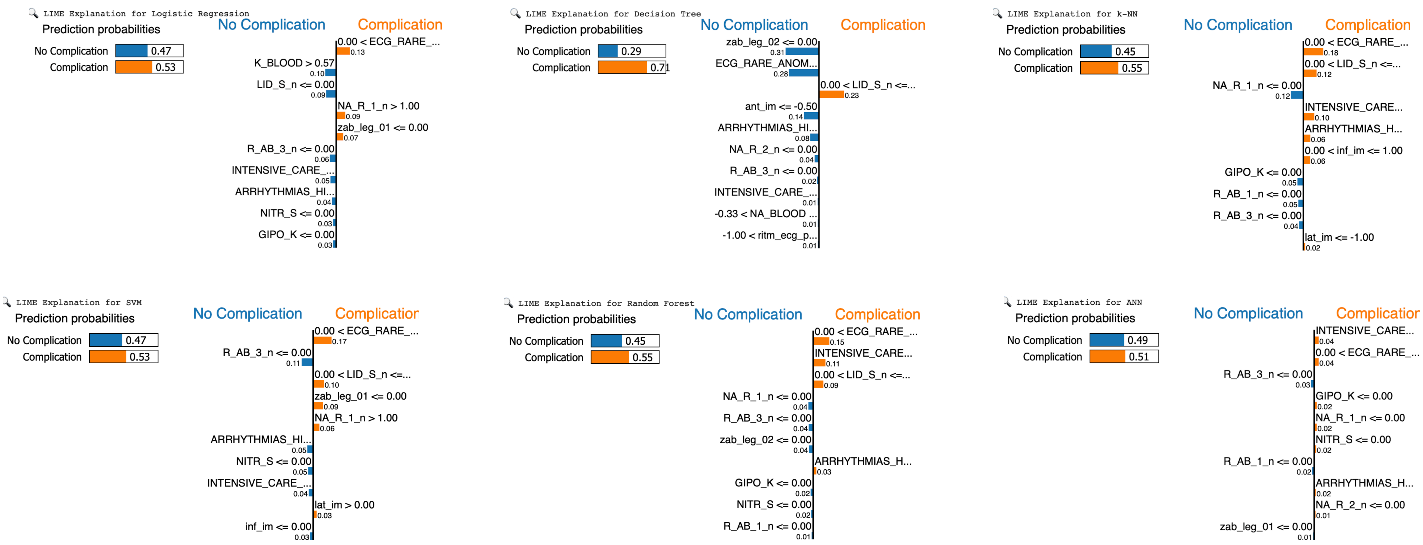
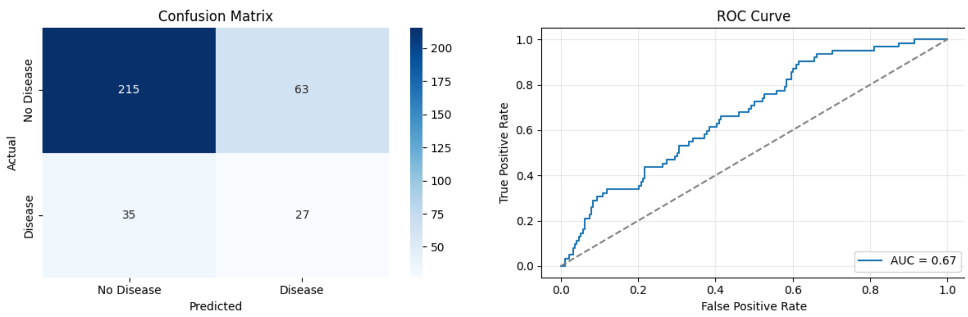


Figure 6. Local feature contributions for individual predictions using LIME

LIME plots for different classifiers highlight the top features influencing the prediction for a representative high-risk patient. Across models, rare ECG anomalies and lidocaine use recurrently contributed to positive classification, supporting their role as early arrhythmic indicators.

The ensemble soft-voting classifier, combining all models, produced (Figure 7):

- Accuracy: 71.0%
- AUC: 0.67
- Macro F1-score: 0.36
- Recall (Positive Class): 44%
- Precision (Positive Class): 31%



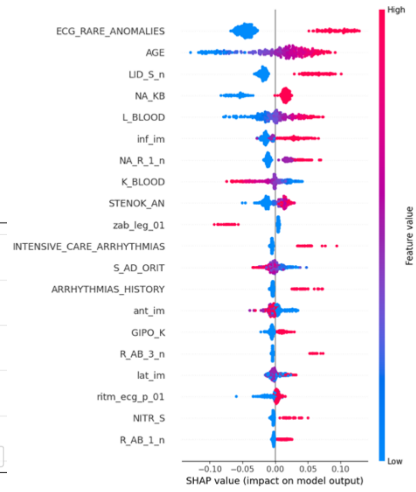


Figure 7. Performance and explainability of the ensemble soft-voting classifier
*This figure summarizes the final performance of the ensemble model on the test set. The confusion matrix (left) shows classification counts; the ROC curve (center) indicates moderate discrimination with an AUC of 0.67; and the SHAP summary plot (right) displays the most influential features, with rare ECG anomalies, age, and lidocaine use (*UD_S_n*) having the greatest impact on model predictions.*

Discussion

The main finding of this study is that multiple machine learning models, ranging from simple interpretable algorithms to more complex ones, achieved comparable and moderate performance in predicting in-hospital arrhythmic complications after AMI. This suggests that, under the current data constraints, model performance is likely limited more by the features available than by model complexity.

The most influential predictors identified across models were rare ECG anomalies at admission, patient age, and lidocaine use by the Emergency Cardiology Team. Clinically, this is coherent: rare anomalies such as ventricular extrasystoles or QT prolongation often indicate myocardial electrical instability; older age is a known risk factor for both supraventricular and ventricular arrhythmias; and the administration of lidocaine in emergency settings is typically reserved for acute ventricular arrhythmias—thus indirectly signaling an early arrhythmic phenotype. Other relevant features included markers of electrolyte imbalance (e.g., sodium-related therapies) and signs of systemic stress or inflammation, supporting the multifactorial nature of arrhythmic complications in AMI patients.

Nevertheless, the study has several limitations. First, the dataset was collected in a single-center, retrospective context, which may limit generalizability. Second, certain key risk factors—such as infarct size, troponin dynamics, and echocardiographic parameters—were not available. Moreover, the class imbalance constrained the achievable precision, despite SMOTE application.

Still, even with moderate predictive performance, the models may serve as clinical decision-support tools to flag high-risk patients early during hospitalization, guiding intensified monitoring or preventive interventions. The model can be seen as a dynamic risk stratification tool, potentially complementary to clinical judgment.

Conclusions

This study assessed whether machine learning models can predict arrhythmic complications in AMI patients using routine clinical and ECG data. Despite using diverse algorithms, all models showed similar and moderate performance, with F1-scores between 0.36 and 0.41 and AUC values around 0.65–0.69. These findings suggest that, given the current dataset, data quality and feature informativeness represent the main limitations, rather than model architecture. Still, the consistent identification of clinically meaningful predictors—like age, rare ECG anomalies, and lidocaine use—supports the validity of the approach. While not ready for autonomous decisions, these models may aid early risk stratification, helping clinicians prioritize high-risk patients for continuous monitoring or further diagnostic evaluation. Explainability methods further enhance clinical applicability. Future research should focus on external validation and inclusion of richer clinical and imaging data.

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